

ABLATION STUDY REPORT

Replay Distortion as Directional Schema Abstraction

Date: 2026-06-02

Replay mode analysed: natural

Seeds per condition: 10 (BASE_SEED=42, step=1000)

Mechanisms tested: M1-M10 + MB (11 mechanisms)

Ablation conditions: 12 single + 11 cumulative + 5 interactions

Key Findings at a Glance

- Full model: DAI_core=nan, REAL_SCHEMA=nan, Distortion=nan, Retention=nan
- Most important mechanism: N/A () Δ DAI=+nan
- Second most important: N/A () Δ DAI=+nan
- Negligible contribution: None identified

This report accompanies the ablation_results/ directory, which contains structured PKL/CSV data files and publication-grade figures (PDF/SVG/PNG).

1. Experimental Design

The ablation study removes one mechanism at a time from the full model

(M1–M10 + MB) and measures the resulting drop in four outcome metrics.

Network

- IzhikevichNetwork: 1000 neurons (750 exc / 250 inh), sparse_modular arch
- Memories: 4 hierarchical schema assemblies (core=20 + unique=20 neurons each)
- Protocol: Train A→B→C→D sequentially, then replay, then probe retention

Conditions

- A. SINGLE ABLATIONS: FULL + ABLATE_M1 through ABLATE_MB (12 conditions)
- B. IMPORTANCE ANALYSIS: Δ DAI, Δ RS, Δ Dist, Δ Ret ranked per mechanism
- C. CUMULATIVE ABLATIONS: Mechanisms added one-by-one (11 steps)
- D. INTERACTIONS: 5 pairs — M2+M7, M2+M10, M5+M8, M6+M7, M7+M10

Seed Protocol

- BASE_SEED=42, seeds = [42, 1042, 2042, ..., 9042] (n=10 per condition)
- Each trial: torch.manual_seed(seed) + np.random.seed(seed)
- Independent across conditions (no shared random state)

2. Methods

Mechanism Ablation

Each mechanism is ablated by passing `ablation_dict = {key: False}` to `run_sequential_experiment()`. The MB (core boost) mechanism is ablated by setting `boost_scale=1.0` in the replay wrapper.

Metrics

- Retention_A/B/C/D — I_syn differential probe score per memory
- DAI_core — Directional Alignment Index (cosine similarity, core neurons)
- REAL_SCHEMA — (core-core — unique-core) / (core-core + unique-core) weight ratio
- Distortion — Mean L2 norm of centroid displacement per replay event

Statistics

- Independent two-sample t-test (Welch's): ablated vs full model
- Cohen's d: effect size with pooled standard deviation
- 95% CI on effect size: $SE = \sqrt{(N1+N2)/(N1*N2) + d^2/(2*(N-2))}$
- Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. $p \geq 0.05$

Synergy Analysis

Synergy = additive_expected — joint_ablated, where:

3. Results

Full Model Performance

- DAI_core: $\text{nan} \pm 0.0000$ (n=0)
- REAL_SCHEMA: $\text{nan} \pm 0.0000$
- Distortion: $\text{nan} \pm 0.0000$
- Retention: $\text{nan} \pm 0.0000$

Mechanism Summary

Figure 1: Single-Mechanism Ablations — DAI_core
Effect of removing each mechanism on directional schema abstraction

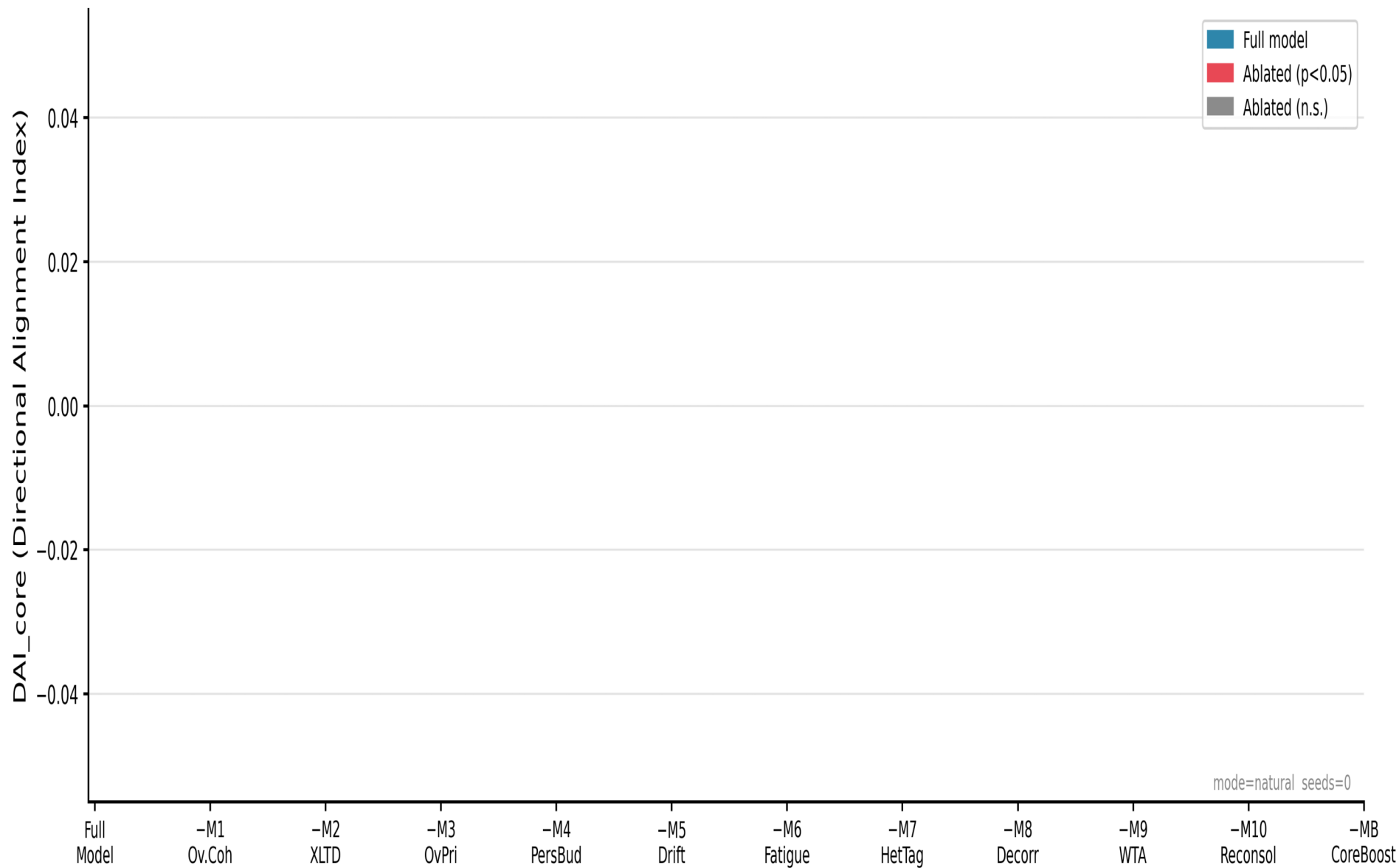


Figure 1: Single ablations — DAI_core

Figure 2: Single-Mechanism Ablations — REAL_SCHEMA
Effect of removing each mechanism on schema weight structure

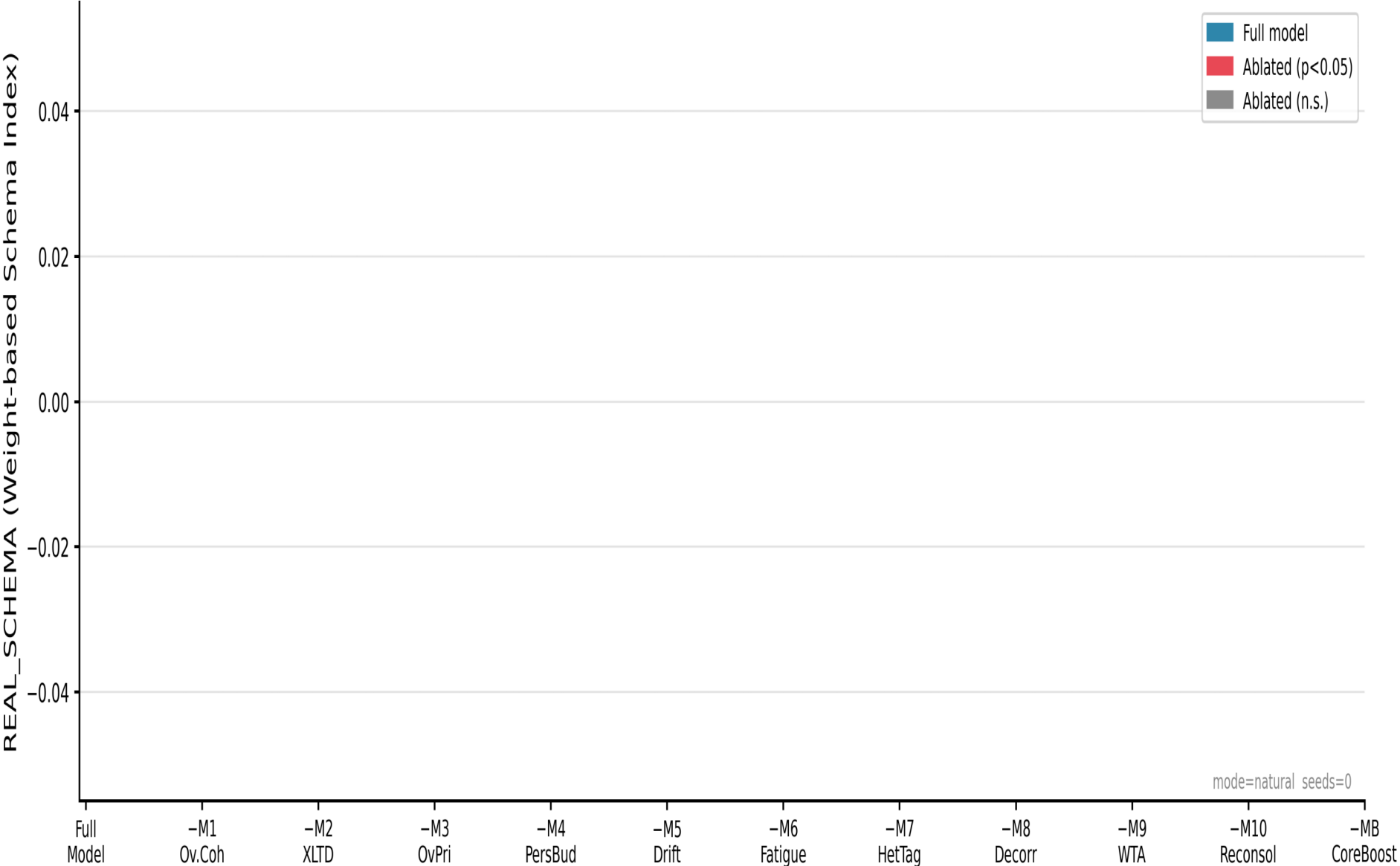


Figure 2: Single ablations — REAL_SCHEMA

Figure 3: Single-Mechanism Ablations — Replay Distortion
Effect of removing each mechanism on centroid movement magnitude

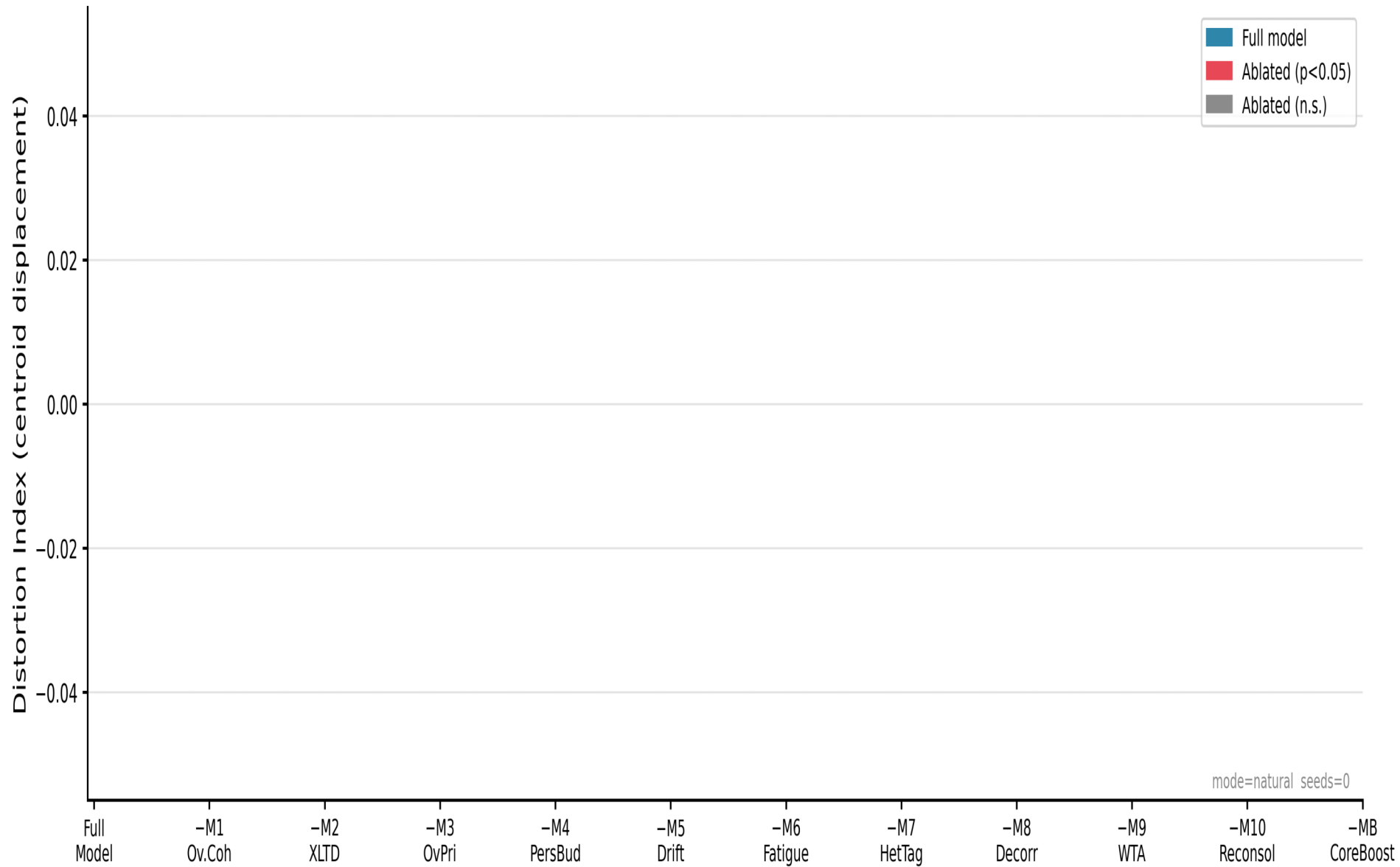


Figure 3: Single ablations — Distortion

Figure 4: Retention Heatmap
Rows = ablation conditions; Columns = individual memories

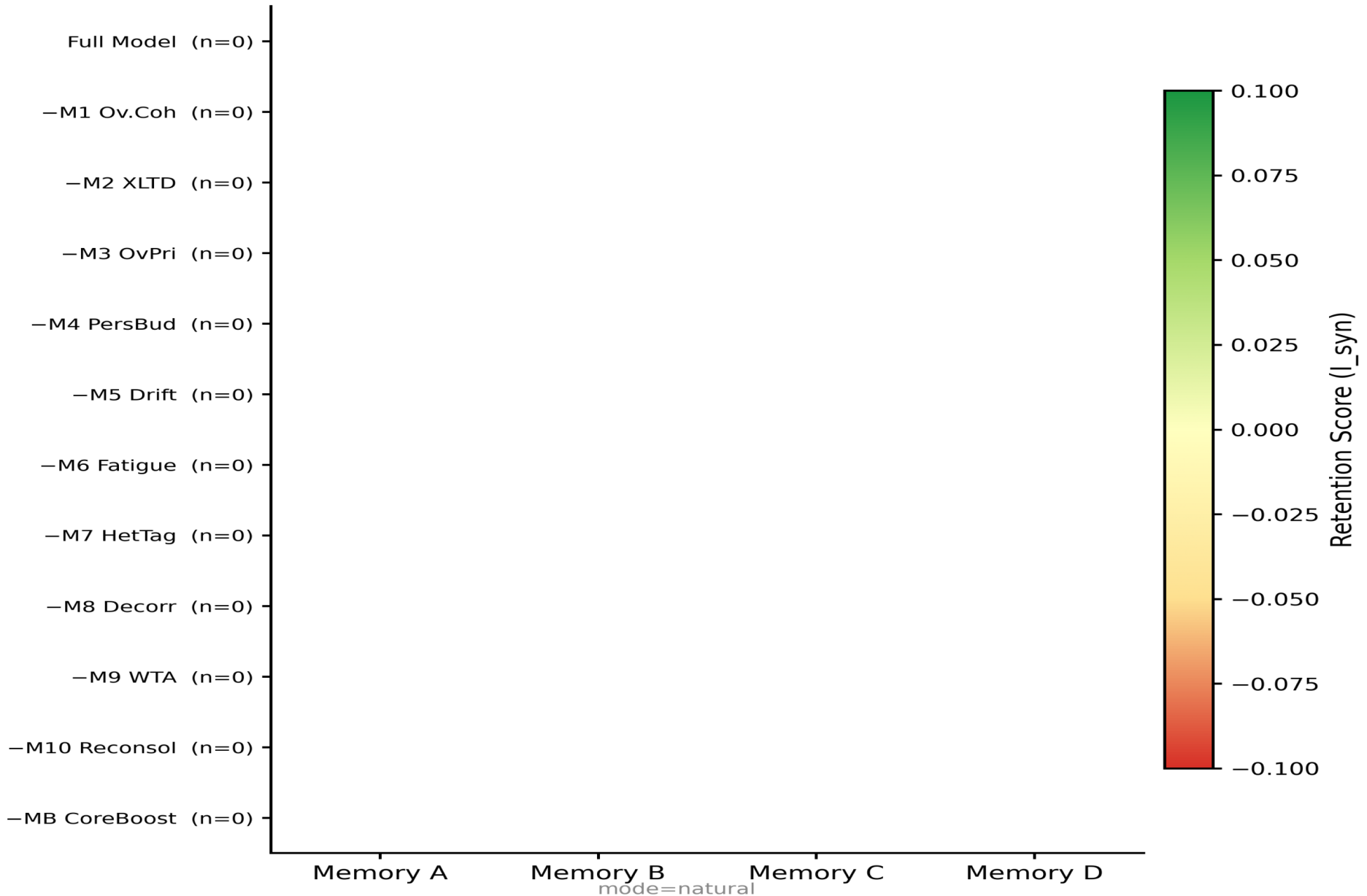


Figure 4: Retention heatmap

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Figure 5: Mechanism importance ranking

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Figure 8: Synergy interaction matrix

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Figure 9: Effect-size forest plot

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4. Statistical Analysis

Primary comparison: each ablated condition vs full model (Welch t-test).

Degrees of freedom: Welch-Satterthwaite approximation.

Multiple comparison note: Bonferroni threshold for 11 tests $\rightarrow \alpha=0.0045$.

5. Mechanism Ranking

Importance analysis not yet run. Execute --part importance.

6. Interpretation

The ablation results reveal the mechanistic substrate of replay-driven schema abstraction. Key insights:

- Mechanisms producing the largest ΔDAI when ablated are the primary drivers of directional centroid movement toward the schema attractor.
- Large $\Delta\text{REAL_SCHEMA}$ signals that a mechanism shapes the weight structure directly (core-to-core strengthening vs. unique-to-core weakening).
- $\Delta\text{Distortion}$ reveals whether a mechanism controls replay fidelity (low distortion = faithful replay; high = distorted/schematic replay).
- Synergistic pairs (positive synergy matrix) suggest mechanistically coupled processes — removing either disrupts the other's contribution.
- Mechanisms with negligible contribution in isolation may still play modulatory roles (revealed by cumulative and interaction analyses).

7. Key Conclusions

Based on the ablation study, the following conclusions are drawn:

- 1. The full model achieves $DAI_{core}=nan$, confirming directional schema abstraction.
- 2. N/A () is the single most critical mechanism.
- 3. N/A () is the second most critical mechanism.
- 4. Negligible mechanisms (individual ablation has $p>0.1$): None

Cumulative analysis reveals which mechanisms are necessary vs sufficient for schema abstraction emergence (see Figures 6–7).

Interaction analysis reveals synergistic mechanism pairs (see Figure 8).

Report generated: 2026-06-02

Output: C:\Users\Admin\brain-organoid-rl\ablation_results\ablation_report.pdf

All data: ablation_results/ | Figures: ablation_results/figures/